# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

## Ch 5 Test: Life Processes

*Chapter / Topic: Life Processes*

Name: _____ Roll No.: _____ Date: _____

**Time: 50 minutes** **Max. Marks: 25**

---

**General instructions:**

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled diagrams wherever required.

### SECTION A

6 marks

*Choose the correct option.*

**Q1.** The kidneys in human beings are a part of the system for: **[1]**

(a) nutrition (b) respiration
(c) excretion (d) transportation

**Q2.** The xylem in plants is responsible for the transport of: **[1]**

(a) food (b) water
(c) amino acids (d) oxygen

**Q3.** The autotrophic mode of nutrition requires: **[1]**

(a) carbon dioxide and water (b) chlorophyll
(c) sunlight (d) all of the above

**Q4.** In aerobic respiration, the breakdown of pyruvate to give carbon dioxide, water and energy takes place in the: **[1]**

(a) cytoplasm (b) chloroplast
(c) nucleus (d) mitochondria

**Q5.** Which blood vessel carries oxygenated blood from the lungs to the heart? **[1]**

(a) Pulmonary artery (b) Aorta
(c) Pulmonary vein (d) Vena cava

**Q6.** **Assertion (A):** The walls of the ventricles are thicker than the walls of the atria. **Reason (R):** The ventricles have to pump blood out of the heart into various organs with high pressure. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

### SECTION B

4 marks

**Q7.** How are the alveoli designed to maximise the exchange of gases? **[2]**

**OR**

Name the three substances secreted by the gastric glands, and give the function of each.

**Q8.** Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds? **[2]**

## SECTION C

6 marks

**Q9.** Answer the following about photosynthesis: **[3]**

**(i)** Write the balanced equation for photosynthesis. **[1]**

**(ii)** List the three events that occur during photosynthesis. **[2]**

**Q10.** Describe how fats are digested in our body. **[3]**

**OR**

Describe the structure and working of a nephron.

## SECTION D

5 marks

**Q11.** Answer the following: **[5]**

**(i)** Give three differences between aerobic and anaerobic respiration. **[3]**

**(ii)** Why do we sometimes get muscle cramps after vigorous exercise? **[2]**

**OR**

(a) What is translocation in plants? Which tissue carries it out? [2]
(b) How does water move upward in tall plants? [3]

## SECTION E

4 marks

*Read the passage and answer the questions that follow.*

**Q12.** When the kidneys fail, an artificial kidney (dialysis machine) can be used. It contains a number of tubes with a semi-permeable lining, suspended in a tank filled with dialysing fluid. This fluid has the same osmotic pressure as blood, except that it contains no nitrogenous wastes. The patient's blood is passed through these tubes, and the nitrogenous wastes pass from the blood into the dialysing fluid. The purified blood is pumped back into the patient. **[4]**

**(i)** Name the main nitrogenous waste removed from the blood. **[1]**

**(ii)** Why does the dialysing fluid contain no nitrogenous wastes? **[1]**

**(iii)** How is dialysis different from the working of a healthy kidney? **[2]**

*~ All the best! ~*